# Aiden Zhang

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## Education

**University of Waterloo -** *BASc in Honours Computer Engineering with Co-op* Sept 2026 - Apr 2031

## Technical Skills

**Languages:** Python, Java, C++, SQL, HTML, CSS, TypeScript, JavaScript, C, Go
**Frameworks:** Spring Boot, Spring MVC, Hibernate/JPA, pandas, NumPy, React.js, PyTorch, TensorFlow, OpenCV, scikit-learn, YOLO, Keras
**Developer Tools:** Git, GitHub, Bitbucket, VSCode, Docker, Maven, AWS, Azure, GCP, Databricks, Vite
**Databases & Technologies:** PostgreSQL, Oracle, SQL Server, MongoDB, REST APIs, ROS 2, Arduino, I2C, UART, RAG

## Experience

**Full-Stack Software Engineering Intern -** MJVA Technology Ltd.
*Java, Go, Spring Boot, REST APIs, Oracle, SQL Server, Maven, CI/CD, JUnit* Jan 2026 - Aug 2026

- Developed secure RESTful APIs using Java 17/21 and Spring Boot to integrate enterprise applications with Moneris financial gateways
- Optimized SQL/PL/SQL queries, stored procedures, and triggers across Oracle and SQL Server processing structured JSON/XML data
- Configured Maven build workflows and automated CI/CD deployment pipelines, validating backend services, RESTful API functionality, and application reliability using JUnit and Postman across multiple development and testing environments
- Managed Git and Bitbucket repositories, maintaining branch consistency through feature branches, pull requests, code reviews, debugging, and Agile Scrum development workflows to support code quality and integration across enterprise development projects

**Autonomy Software Team Member -** Waterloo Aerial Robotics Group (WARG) Sept 2026 - Present

- Build software for autonomous drone operations, including behaviour-tree telemetry, emergency flight safety, and multi-agent coordination

**EVE Software Team Member -** WATonomous Sept 2026 - Present

- Work on autonomous vehicle perception for WATonomous' flagship vehicle, multi-object tracking, and robotic navigation using ROS 2

## Projects

**Computer Vision Food Recognition and Nutritional Analysis System**
*Python, PyTorch, YOLO, OpenCV, NumPy, pandas, TensorFlow* Jun 2025 - Oct 2026

- Built a real-time food detection app that finds and identifies 5 food categories in live webcam video, replacing an earlier TensorFlow classifier (96% test accuracy) with a fine-tuned YOLO11 model that boxes each food and detects several items at once
- Trained the model on **245 labeled images**, reaching **98% mAP50** on a validation set and averaging **~13.5 FPS across 300 frames**
- Ran a 5-month pilot with 6 users, with a visually impaired participant who used TTS to access nutrition feedback supporting nutrition-aware choices; participants reported healthier eating habits

**Drone Behaviour Tree Telemetry System - WARG**
*Python, ROS 2, MAVLink, MAVROS, py_trees, Docker, ArduPilot SITL, Git* Sept 2026 - Nov 2026

- Engineered a ROS 2 behaviour tree telemetry system to provide ground operations with autonomy state visibility without LTE connectivity, encoding active behaviours into compact MAVLink STATUSTEXT messages with **≤ 50-byte payloads**
- Implemented automated behaviour-to-ID mapping and parent-child relationship tracking from live ROS 2 snapshots, using efficient bandwidth identifiers, and two transmission mechanisms (event-driven state updates and **30-second periodic retransmission**)
- Validated 4 autonomy states (Takeoff, FlyToWaypoint, RepeatLap, and WaitForReconComplete) with end-to-end integration testing using Docker, ArduPilot SITL, MAVROS, and Mission Planner confirming successful MAVLink STATUSTEXT reception to ground station

**Recall - AI Study Assistant Chrome Extension**
*Java 21, Spring Boot, React, TypeScript, PostgreSQL, SQL, Vite, CSS, Git, npm* Oct 2026 - Present

- Built a Chrome Manifest V3 side-panel extension with React and TypeScript that generates fill quizzes and prediction papers
- Designed a Java 21 and Spring Boot REST API backed by PostgreSQL that persists quizzes, questions, and attempts across sessions

**Autonomous Robot Navigation and LiDAR Mapping - WATonomous**
*C++, ROS 2, Foxglove, Docker* Sept 2026

- Built a 4-node C++/ROS 2 navigation system integrating LiDAR mapping, odometry fusion, A* pathfinding, and 10 Hz Pure Pursuit
- Engineered **100 x 100 m occupancy grids** with **0.1 m resolution**, **0.65 m obstacle inflation**, and **8-direction A* planning**

**Bedside Patient Monitoring System**
*Arduino Uno, C++, Embedded Programming, I2C, UART, Serial Communication, ADC* Oct 2026 - Present

- Developed an embedded monitoring prototype integrating heart-rate sensing and **3-axis acceleration measurements** to track pulse rate